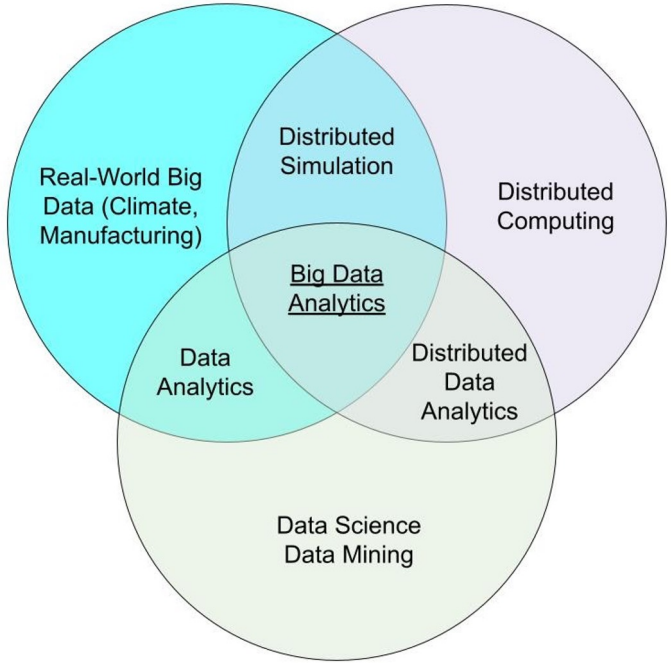
Bathymetry SR-ResNet: Enhanced Deep Learning Based Super-Resolution for Bathymetry Data

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Research Portfolio

- Centered on **big data analytics** with three connected and supporting areas
  - **Distributed computing**: key enabling computing infrastructure for big data
  - **Data science**: core research topic to learn patterns from big data
  - **Real-world big data**: research frontier to identify new research challenges and evaluate research contribution and impacts
- Current research topics
  - Climate Informatics
  - Causality Analytics
  - Streaming Data Analytics

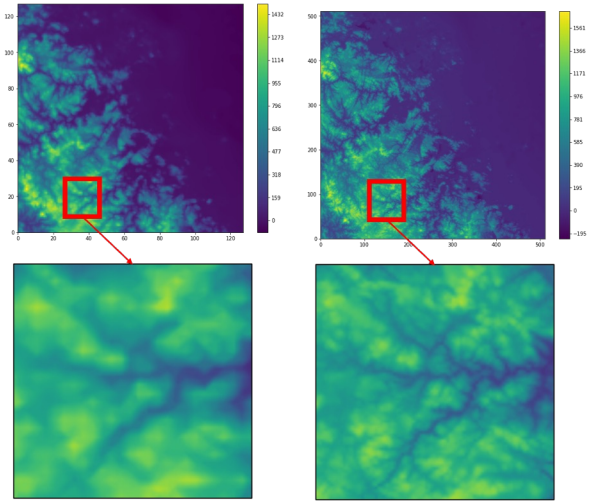


Background

Bathymetry is an important measurement of **water depth**

- Essential for thematic processing of remote sensing data
- Essential for physical and optical satellite observations
- Helps in the physical measurements, such as land surface temperature, aerosol retrievals, cloud detection, etc. [1]
- Improve the quality of environmental products

However high-resolution (HR) bathymetry is not always available due to limitations on equipment.



Colormaps of cropped images from two different bathymetry dataset. The spatial resolution of ETOPO-1 (left) is 4 times of GEBCO (right).

Related Work

Single Image Super-Resolution (SISR) is a solution to reconstructing high-resolution (HR) images from low-resolution (LR) images.

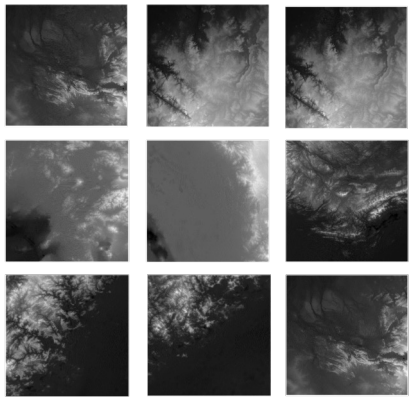
- Deep learning-based methods has been proved highly accurate [2, 3, 4]
- Recover photo-realistic and natural textures well
- Trained and test using benchmarking datasets of **RGB** images, which are different from single-band **bathymetry data**

Existing works of SISR for Bathymetry Data [5]

- Lacking in deep learning based approaches or improvements
- Lacking in conventional evaluation metrics specialized for bathymetry data
- Large bias might exist: Some of the studies are local-focused, which means the model is tested partially or only on images representing the same areas used as training data.

Therefore, our work is to enhance an **deep learning** based SISR model, **SR-ResNet** [3], to generate HR images for bathymetry data.

Challenges



Bathymetry images
Pixel value: 16-bit, integer/float



RGB images
Pixel value: 8-bit, integer

- Most of the current SISR models are constructed based on benchmark datasets consisting of natural pictures with **RGB** bands, while bathymetry is **single-band grey-scaled.**
- **Bathymetry data amount is limited**.
- **Coastal area is in our interest** and requires more accurate measurements than other areas.

Bathymetry Dataset

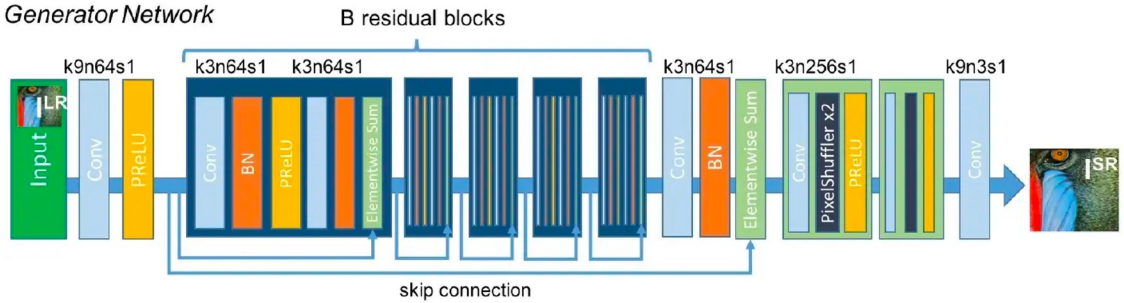
- 1,600 training tiles, 400 test tiles
- Manually cropped and selected from **GEBCO** [6] and **ETOPO-1** [7] **global grid**.
- Each tile includes one high resolution tile from GEBCO and one low-resolution tile from ETOPO.
- Each tile represent 1.016 decimal degree boxes, equivalent to exactly **512x512** pixel areas in the GEBCO data and **128x128** pixels in ETOPO-1.

Comparison between ETOPO-1 and GEBCO dataset

Data	Source	Spatial Resolution
ETOPO-1	NOAA National Centers for Environmental Information	1 arc-minute
GEBCO_2021 Grid	General Bathymetric Chart of the Oceans (GEBCO) with support from the International Hydrographic Organization (IHO) and Intergovernmental Oceanographic Commission (IOC)	15 arc-seconds

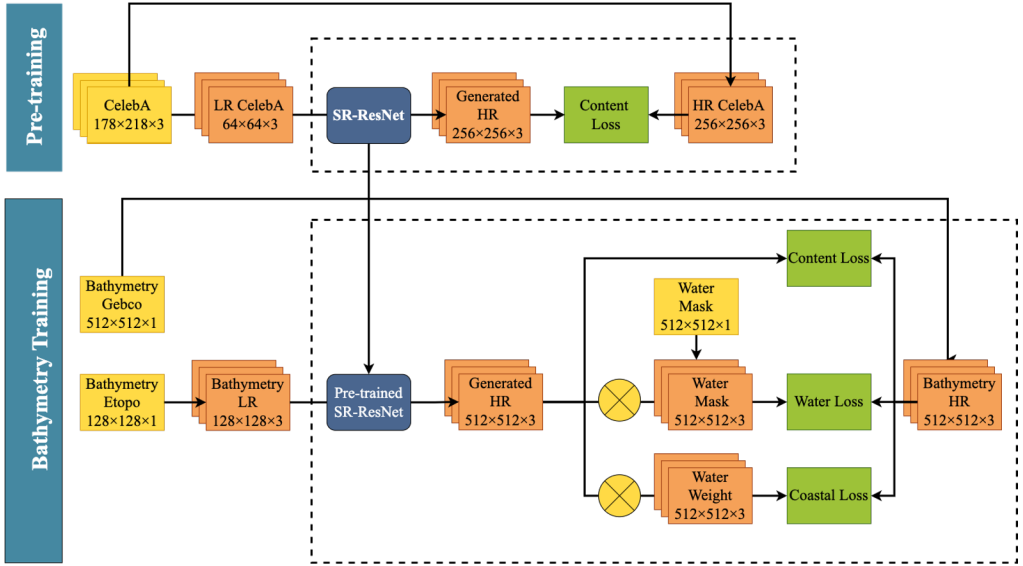
SR-ResNet

- SR-ResNet is a generative network for SISR, with an upscaling factor 4 on image height and width.
- SR-ResNet [4] was proposed by Ledig et al. The paper was published in 2017 CVPR.
- The measurements on benchmark data *Set5*: PSNR = 32.05, SSIM = 0.9019.
- There are B **residual blocks** (B=16) in the network originated by ResNet [9].



Architecture of SR-ResNet[4] with corresponding kernel size (k), number of feature maps (n) and stride (s) indicated for

Bathymetry SR-ResNet



Overall architecture of the bathymetry super-resolution model. In **pre-training**, a benchmark RGB-banded dataset CelebA [8] is utilized to train a SR-ResNet from the start with content loss. In **bathymetry training**, bathymetry datasets is pre-processed and then used to train SR-ResNet based on the pre-trained model with knowledge-based loss functions (water loss and weighted coastal loss). The trained model is saved to generate HR images for any LR inputs with an upscaling factor **4** .

Transfer Learning

- Transfer learning is used to apply a machine learning technique that focuses on learning knowledge from one domain (task) and applying it to a different but related domain (task).
- In our proposed framework, the structure and relationship between low and high-resolution images used in large-scale training can be learned by deep convolutional neural networks (SR-ResNet) and be transferred to Bathymetry images:
    - First, the SR-ResNet network is pre-trained on a Large-scale CelebFaces Attributes (CelebA) dataset containing 202,599 JPEG images
    - Second, the pre-trained SR-ResNet with updated parameters is used as initial model to give the model some 'knowledge' of training, including backbone structure and mapping
    - Next, we enhance the resolution of Bathymetry images by transferring the knowledge from the readily available dataset CelebA with a different task via Bathymetry training

4.3 Loss Functions - SR-ResNet

- SR-ResNet was proposed by Ledig et al. [3] with SRGAN.
- Loss function: Content Loss
  - First, extract features from images using a VGG network
  - The loss compares differences between feature maps of the generated HR image and feature maps of the true HR image
  - **Content Loss = VGG-based loss**

$$l_{VGG} = \frac{1}{W_{i,j} H_{i,j}} \sum_{x=1}^{W_{i,j}} \sum_{y=1}^{H_{i,j}} (\phi_{i,j}(I^{HR})_{x,y} - \phi_{i,j}(G_{\theta_G}(I^{LR}))_{x,y})^2$$

- $W_{i,j}, H_{i,j}$: Width and height of feature map obtained by the j-th convolution before the i-th max pooling layer within the VGG19 network
- x, y: The pixel at x-th column and y-th row
- $\phi_{i,j}(I^{HR})$: Feature map of HR image
- $\phi_{i,j}(G_{\theta_G}(I^{LR}))$: Feature map of generated image

- In our architecture, the original SR-ResNet is used only for pre-training

Loss Functions in Bathymetry SR-ResNet

- Bathymetry SR-ResNet is our improvement on SR-ResNet for bathymetry data
- Loss function: Content Loss + knowledge-based loss functions
  - **All loss functions** equal sum of **VGG-based loss + Pixel-based loss**
  - Knowledge-based loss functions: **loss_water, loss_weighted**
    - loss_water: use binary masks E to focus on water area
    - loss_weighted: compute a weight mask $M_{i,j} = \frac{1}{||I^{HR}||} \cdot E_{i,j}$ applying higher weight to coastal areas

$$l_{MSE_{water}} = \frac{1}{r^2WH} \sum_{x=1}^{rW} \sum_{y=1}^{rH} (E \cdot I_{x,y}^{HR} - E \cdot G_{\theta_G}(I^{LR})_{x,y})^2$$

Upscaling factor. r = 4

$$l_{MSE_{weighted}} = \frac{1}{r^2WH} \sum_{x=1}^{rW} \sum_{y=1}^{rH} M_{i,j} \cdot E_{i,j} (I_{x,y}^{HR} - G_{\theta_G}(I^{LR})_{x,y})$$

$$l_{VGG_{water}} = \frac{1}{W_{i,j}H_{i,j}} \sum_{x=1}^{W_{i,j}} \sum_{y=1}^{H_{i,j}} \phi_{i,j}((\phi_{i,j}(E \cdot I^{HR})_{x,y} - \phi_{i,j}(E \cdot G_{\theta_G}(I^{LR}))_{x,y})^2$$

$$l_{VGG_{weighted}} = \frac{1}{W_{i,j}H_{i,j}} \sum_{x=1}^{W_{i,j}} \sum_{y=1}^{H_{i,j}} \phi_{i,j}(M_{i,j} \cdot E_{i,j})_{x,y} \cdot (\phi_{i,j}(I^{HR})_{x,y} - \phi_{i,j}(G_{\theta_G}(I^{LR}))_{x,y})^2$$

Experiment Metrics

Evaluation metrics: MSE, SSIM, PSNR, WPR

- Mean Squared Error (MSE): pixel-level differences between generated images and ground truth. ↓

$$MSE = \frac{1}{m\,n}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[I(i,j) - K(i,j)]^2.$$

- Structural Similarity Index Measure (SSIM): the similarity between two images on luminance $l(x,y)$, contrast $c(x,y)$, and structure $s(x,y)$. ↑

$$\mathrm{SSIM}(x,y) = \left[l(x,y)^{\alpha} \cdot c(x,y)^{\beta} \cdot s(x,y)^{\gamma}\right]$$

- Peak Signal-to-Noise Ratio (PSNR): the ratio between the maximum possible power of a signal and the power of corrupting noise. ↑

$$PSNR = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

- Wrong pixel ratio (WPR): the percentage of generated pixels with an opposite sign to the ground truth. The wrong sign indicates the pixel mistake between land area and water area. ↓

Models used in experiment

Baseline models:

- Baseline 1: bi-cubic interpolation
- Baseline 2: SR-ResNet trained by CelebA dataset without transfer learning
- Baseline 3: SR-ResNet trained by bathymetry dataset without transfer learning

Experiment models:

The experiment models follow the architecture of Bathymetry SR-ResNet, which is trained starting from a pre-trained SR-ResNet. Different models utilize different combinations of loss functions:

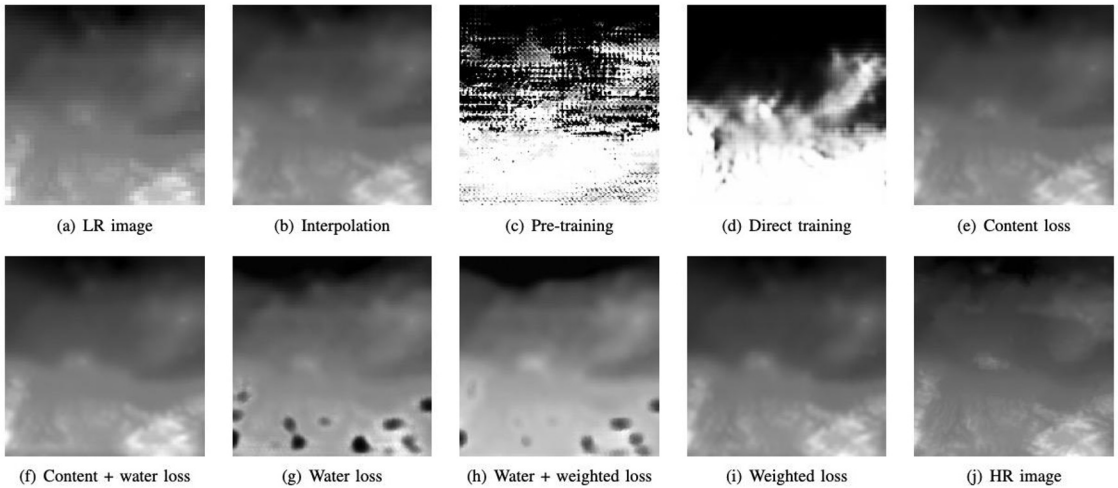
- Model 1: Bathymetry SR-ResNet trained with content loss
- Model 2: Bathymetry SR-ResNet trained with content loss + water loss
- Model 3: Bathymetry SR-ResNet trained with water loss
- Model 4: Bathymetry SR-ResNet trained with water loss + weighted loss
- Model 5: Bathymetry SR-ResNet trained with weighted loss

Experiment Results

Three baseline models and five experiment models with corresponding evaluation results of MSE, PSNR and SSIM over the **whole image**, **ocean area** and **coastal area**, respectively.

	Evaluation on whole area			Evaluation on ocean area			Evaluation on coastal area			
Models	**MSE**	**PSNR**	**SSIM**	**MSE**	**PSNR**	**SSIM**	**MSE**	**PSNR**	**SSIM**	**WPR**
Bi-cubic interpolation	**1,303**	**49.654**	**0.988**	**7,853**	**47.401**	**0.989**	**1,771**	**52.996**	**0.995**	**0.011**
Pre-training	58,574,684	-0.846	0.004	57,881,520	-0.7663	0.220	37,101,514	1.817	0.481	0.284
Direct training	59,753,964	-1.027	0.023	60,407,267	-1.0263	0.240	40,019,392	1.440	0.484	0.302
Content loss	**62,357**	34.096	0.782	**62,137**	34.136	0.808	24,248	37.825	0.867	0.135
Content + water loss	612,187	**38.994**	**0.927**	611,307	**39.613**	**0.946**	31,449	**44.752**	**0.978**	**0.084**
Water loss	4,081,509	13.841	0.236	3,813,084	14.786	0.511	2,090,382	17.871	0.621	0.354
Water + weighted loss	3,310,707	14.3676	0.264	2,992,009	15.448	0.544	1,589,616	18.736	0.642	0.347
Weighted loss	**142,761**	36.039	0.898	102,043	37.294	**0.940**	**21,555**	**42.317**	**0.963**	0.133

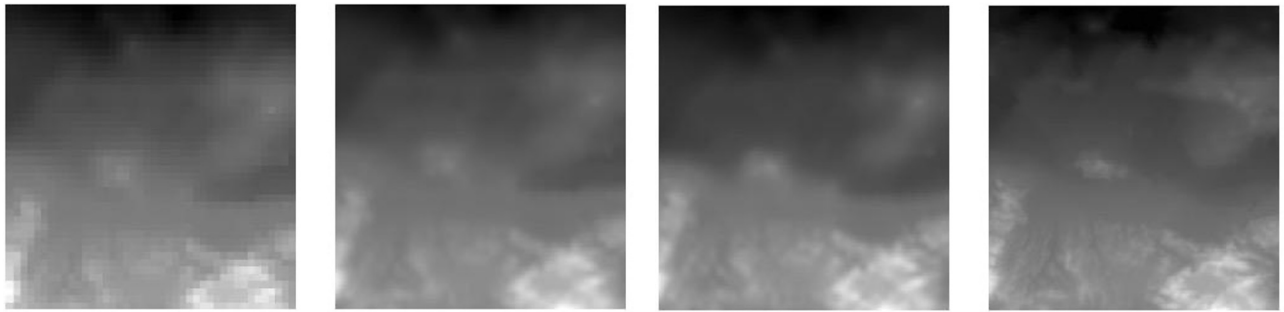
Experiment Results (2)



The visual comparison among different experiments and ground truth at the same location. From (a) to (j) are original low-resolution image, bi-cubic interpolation (baseline 1), pre-trained model (baseline 2), directly-trained model (baseline 2), model 1 (content loss), model 2 (content + water loss), model 3 (water loss). model 4 (water + weighted loss), model 5 (weighted loss) and original high-resolution image (ground truth). Among all the experiments, **model 5 (weighted loss) in sub-image (i)** is the best model we have. The area in **white color has positive pixel values** and represents the land area, the **black area represents the deep ocean** and the area in grey around land area represents the coastal area.

Experiment Results (3)

- Quantitative evaluation results shows the interpolation method performs the best in terms of MSE, PSNR, SSIM and WPR.
- Qualitative evaluation (visual comparison) shows that **our model using knowledge-based loss functions shows better perceptual features**.



(a) LR image (b) Interpolation (i) Weighted loss (j) HR image

Conclusions and Future Work

Conclusions:

- The Bathymetry Super-Resolution model reconstruct the ocean bathymetric data by enhancing the SR-ResNet-based model trained by GEBCO and ETOPO-1 dataset.
- We enhance the SR-ResNet by knowledge-based loss functions (water loss and weighted loss) and transfer learning to train the model with limited bathymetry data.
- The results show that the Bathymetry SR-ResNet generates high-spatial-resolution bathymetry that recovers more accurate perceptual features in the coastal area.
- The quantitative results show that only MSE, PSNR and SSIM do not tell the quality of generated images, which is the same as the conclusion of [4].

Future work:

- Find reliable evaluation metrics for bathymetry super-resolution.
- Enlarge the dataset by collecting more tiles for training and testing.

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Questions?

- GitHub: https://github.com/big-data-lab-umbc/bathymetry_super_resolution
- Website: https://bdal.umbc.edu/
- Email: Jianwu@umbc.edu